

TIME MACHINE

3D scroll storytelling implementation plan

A production blueprint for a cinematic historical timeline inspired by the interaction language of The Year of Greta - without copying its subject matter or implementation wholesale.

DESIGN INTENT

A dark, archival-futurist experience where a central time-machine object anchors a spatial timeline. As the visitor scrolls, notable event cards orbit, unfold and approach the viewer; clicking an event opens a deeper story layer without leaving the scene.

TYPE SYSTEM

Primary: Satoshi Regular / Bold Utility + dates: Roboto Mono Regular / Medium

REFERENCE CHECK

The Year of Greta is documented as an illustrated 2019 timeline using Three.js for the 3D world. Communication Arts describes rotating planes around Greta, vertex displacement for motion/hover, an environment map, smoke particles and a generated noise-texture background. Other references tag the project with GSAP + Three.js.

The Year of Greta: the parts worth borrowing

This page separates verified characteristics of the reference from our proposed architecture.

VERIFIED: PROJECT + FORM

The Year of Greta is an illustrated timeline about Greta Thunberg's rise in 2019, created by Superhero Cheesecake. The project used a 3D model developed with Colorbleed.

VERIFIED: 3D WORLD

Communication Arts states that Three.js was used to create a tornado-like path spinning around Greta. Planes were rotated around her; their vertices were displaced to create motion and hover effects.

VERIFIED: LIGHT + ATMOSPHERE

The team used an environment map for real-life-style lighting and performance benefits, then added smoke particles and a generated noise-texture background that rotated with the path.

REFERENCED BY OTHER DESIGN SOURCES

Design-reference sites tag The Year of Greta with Three.js and GSAP animation. This is useful evidence for the animation category, but the Communication Arts technology note is the stronger source for the specific mechanics above.

SOURCE TRAIL

1. Communication Arts: <https://www.commartarts.com/webpicks/the-year-of-greta>
2. Three.js Forum showcase: <https://discourse.threejs.org/t/the-year-of-greta-thunberg/12380>
3. The Year of Greta: <https://theyearofgreta.com/>
4. Landing.Love reference: <https://www.landing.love/sites/theyearofgreta/>

The recommended production stack

Keep React responsible for UI/content, and keep WebGL responsible for spatial presentation. That separation makes the experience easier to scale, debug and make accessible.

Layer	Choice	Role
App	Next.js + TypeScript	Routing, metadata, deployment structure
3D	Three.js + React Three Fiber	Scene, camera, models, interaction
3D helpers	@react-three/drei	Environment, loaders, controls, utility abstractions
Animation	GSAP + ScrollTrigger	Scroll-scrubbed camera/object/card choreography
Scroll feel	Lenis (optional)	Smooth scrolling / inertial feel
Assets	GLB / GLTF	Web-friendly 3D delivery format
Content	Sanity or Supabase	Events, dates, copy, image references
Media	Cloudinary or Vercel Blob	Image/video delivery, transforms
Deployment	Vercel	Build + edge delivery + preview deployments

CORE RULE

Do not put every interface element into WebGL. Keep text, buttons, navigation, modal content and accessibility-critical UI in the DOM. Use WebGL for the time machine, scene, spatial cards, particles and camera choreography.

No Blender required

The fastest path is to generate or source GLB assets, then optimize them before integrating them into the scene.

GENERATION

Use Meshy or Tripo for text/image-to-3D exploration. Design the time machine as several parts rather than one giant mesh: shell, ring, interior mechanism, portal, small timeline props.

Recommended asset breakdown

- time-machine.glb - primary hero asset, kept visually simple and animation-friendly.
- timeline-ring.glb - reusable geometry for the timeline path.
- event-props/*.glb - small category props for selected events.
- fx textures - noise, grain, smoke, displacement and masks.
- event media - compressed WebP/AVIF images; video only where it materially improves storytelling.

OPTIMIZATION GATE

Before an asset enters the app: reduce polygon count, compress textures, remove hidden geometry, confirm normals/materials, and test the GLB on a mid-range mobile device. Load distant or non-active assets lazily.

Typography implementation

Use the requested production fonts in the website itself: Satoshi for titles/body/UI and Roboto Mono for dates, metadata, counters and technical labels. The provided font specimen shows Satoshi Regular/Bold and Roboto Mono Regular/Medium.

Fontshare documents Satoshi as an Indian Type Foundry family available under the ITF Free Font License; GoogleFonts documents Roboto Mono as the monospaced Roboto family.

Scroll -> approach -> unfold -> click -> reveal

The experience should feel like moving through a physical archive in zero gravity. Scroll is the primary input; click is the depth trigger.

STATE 0 - ENTRY

Time machine is centered and mostly still. Ambient particles establish depth. A single instruction establishes the gesture: SCROLL TO TRAVEL.

STATE 1 - TRAVEL

As scroll progress advances, the camera moves through the time-machine axis. Timeline markers pass the viewer. Event cards enter from off-axis positions and settle into the readable zone.

STATE 2 - FOCUS

The active event card receives stronger contrast and a subtle positional lift. Non-active cards reduce opacity/scale slightly. Avoid excessive motion while text is being read.

STATE 3 - REVEAL

Clicking the active event transitions the camera toward the card. The card expands into a glass information surface while the world remains visible behind it.

STATE 4 - RETURN

Close returns the camera to the timeline state without resetting scroll position. Keyboard Escape should perform the same action.

Suggested scroll choreography

0-15%: hero reveal | 15-75%: timeline travel | 75-95%: denser event rhythm | 95-100%: end-state / explore-more

Make the scene data-driven

Every event should be data, not a one-off component. This is the key to going from 5 events to 50 without rewriting animation logic.

Suggested data shape

```
event = { id, year, date, title, category, summary, body, image, source, position, rotation, scale }
```

Suggested project tree

```
app/ page.tsx components/ scene/TimeMachine.tsx scene/Timeline.tsx scene/EventNode.tsx
scene/Particles.tsx ui/EventPanel.tsx ui/Navigation.tsx lib/ events.ts assets.ts styles/ globals.css
public/ models/ textures/ events/
```

SCENE CONTRACT

The scene receives timeline progress and activeEventId. It does not own the historical copy. UI receives the selected event and renders the detailed text in HTML. This keeps animation and content independently testable.

Animation contract

- Scroll progress -> normalized 0..1 value.
- Progress -> camera position/rotation, timeline offset and event visibility.
- Active event -> focus transform and UI reveal.
- Route/query state optional -> deep-linkable event pages without rebuilding the scene.

Performance is a design feature

The visual quality will be judged as much by responsiveness as by the 3D look. Set performance budgets early.

TARGETS

Desktop should feel fluid at typical 60fps conditions. Mobile should degrade gracefully: fewer particles, simpler post-processing, lower texture resolution, reduced model count and potentially a simplified camera path.

Guardrails

- Lazy-load non-critical GLB assets and heavy media.
- Avoid dozens of independent materials; reuse materials and geometries where possible.
- Keep glass primarily in CSS/DOM where possible; true WebGL transparency can be expensive.
- Use device-aware DPR caps rather than rendering at unrestricted native density.
- Use prefers-reduced-motion to disable camera turbulence and continuous particles.
- Preload only the hero model, first event media and critical UI fonts.

ACCESSIBILITY

Provide a conventional text timeline as a fallback view. Every event needs a keyboard-focusable control, a readable DOM title, visible focus state, Escape-to-close behavior and source attribution. Do not make historical information available only through hover.

MOBILE MODE

Keep the same narrative sequence, but allow the 3D machine to occupy less of the viewport and move cards into a more frontal composition. Touch should replace hover states with tap/focus states.

Build the feeling before the full archive

A staged build reduces risk. Do not start with CMS, 30 events and complex shaders.

Phase	Deliverable	Definition of done
1. Foundation	Next.js + R3F + scene bootstrap	Hero page renders; camera + light + one GLB work
2. Scroll prototype	GSAP/ScrollTrigger timeline	Scroll reliably controls camera and one event
3. Event system	3 data-driven glass cards	Cards enter/leave/focus correctly; click opens detail
4. Art pass	Lighting, particles, textures, typography	The world has a coherent visual language
5. Archive	Real historical content + sources	Every event has verified copy, media and source
6. Responsive	Mobile + reduced motion + keyboard	Core story works without mouse and on touch
7. Optimization	Asset compression + lazy loading	Performance budget passes target devices
8. Launch	Analytics + SEO + deployment	Production build, error monitoring and shareable URLs

MVP DEFINITION

One time machine + one timeline path + three glass event cards + scroll-scrubbed camera + click-to-expand detail panel + responsive/mobile fallback. Once this feels excellent, scale the archive.

FINAL RECOMMENDATION

Start with the visual prototype in isolation. Use a generated GLB hero model, three placeholder events and hard-coded data. Only after the scroll interaction feels cinematic should you introduce the content backend and full event library.

Launch checklist

- Reference mechanics are treated as inspiration, not copied code or content.
- Satoshi Regular/Bold + Roboto Mono Regular/Medium are declared as the web typography system.
- Hero GLB is optimized and tested on desktop + mid-range mobile.
- Scroll controls a stable normalized timeline progress value.
- Cards are data-driven and keyboard/touch accessible.
- Clicking an event opens the detail layer without losing timeline position.
- Historical copy includes a source field and source link.
- Reduced-motion mode removes continuous camera turbulence and particle-heavy effects.
- DPR, texture size, model count and loading behavior have explicit budgets.
- Production build has analytics, error monitoring, SEO metadata and social preview image.

REFERENCE + TOOL SOURCES

The Year of Greta / Communication Arts <https://www.commarts.com/webpicks/the-year-of-greta> Three.js Forum showcase <https://discourse.threejs.org/t/the-year-of-greta-thunberg/12380> The Year of Greta live site <https://theyearofgreta.com/> Landing.Love reference (tags: GSAP, ThreeJS) <https://www.landing.love/sites/theyearofgreta/> Satoshi specimen / Fontshare source <https://studio-tokyo.com/typeface/satoshi> <https://www.fontshare.com/fonts/satoshi> Roboto Mono repository / Google Fonts family <https://github.com/googlefonts/RobotoMono>

DESIGN PRINCIPLE

Borrow the reference's sense of spatial narrative. Do not borrow its subject, copy, imagery or exact scene. The objective is a distinct time-machine experience with the same level of interaction ambition.